"10D_1" is the Case where the six curled dimensions
form a torus, T^6, and 3-space and curled space have individual scale factors.

Notation and Setup:

Starting Metric: $g_{00}=c^2$, $g_{11}=-a[t]^2\,B[r]$, $g_{22}=-r^2\,a[t]^2\,C[r]$,
$g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]$, $g_{44}=-b[t]^2\,p_4^2$, $g_{55}=-b[t]^2\,p_5^2$,
$g_{66}=-b[t]^2\,p_6^2$, $g_{77}=-b[t]^2\,p_7^2$, $g_{88}=-b[t]^2\,p_8^2$, $g_{99}=-b[t]^2\,p_9^2$
a(t) and b(t) are independent and not constant

Corresponding Line Element

$$ds^2 = c^2\,dt^2 - a[t]^2\,B[r]\,dr^2 - r^2\,a[t]^2\,C[r]\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,C[r]\,d\phi^2 - b[t]^2\,p_4^2\,d\psi_4^2 - b[t]^2\,p_5^2\,d\psi_5^2 - b[t]^2\,p_6^2\,d\psi_6^2 - b[t]^2\,p_7^2\,d\psi_7^2 - b[t]^2\,p_8^2\,d\psi_8^2 - b[t]^2\,p_9^2\,d\psi_9^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, **$A'=\partial_r A(t,r)$** and **$\dot{A}=\partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) independent scale factors a and b, (2) $\mathcal{R}(t) = c^2$, and (3) C left free, which gives the FG-GRW form of the solution; setting C=1 gives the GRW form.

a: GRW Equations

We begin with the assumptions:

a(t) and b(t) are independent and non-constant,

$A(t,r) = c^2$,

B(r) is a free parameter whose formula we seek,

C(r) is a free parameter,

and $\mathcal{D}_4[r] = \ldots = \mathcal{D}_9[r] = 1$ (which make the torus Ricci flat).

The initial Lambda field equations are:

R_{00}: $\frac{3\ddot{a}}{a c^2} + \frac{6\ddot{b}}{c^2 b[t]} = \Lambda$

R_{01}: $0 = 0$

R_{11}: $\frac{2\dot{a}^2}{a^2 c^2} + \frac{\ddot{a}}{a c^2} + \frac{6\dot{a}\dot{b}}{a c^2 b[t]} - \frac{2C'}{a^2 r C B} + \frac{(C')^2}{2a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C'B'}{2a^2 C B^2} - \frac{C''}{a^2 C B} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{2\dot{a}^2}{a^2 c^2} + \frac{\ddot{a}}{a c^2} + \frac{6\dot{a}\dot{b}}{a c^2 b[t]} - \frac{2C'}{a^2 r C B} + \frac{B'}{2a^2 r B^2} + \frac{C'B'}{4a^2 C B^2} - \frac{C''}{2a^2 C B} = \Lambda$

R_{33}: $\frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{2\dot{a}^2}{a^2 c^2} + \frac{\ddot{a}}{a c^2} + \frac{6\dot{a}\dot{b}}{a c^2 b[t]} - \frac{2C'}{a^2 r C B} + \frac{B'}{2a^2 r B^2} + \frac{C'B'}{4a^2 C B^2} - \frac{C''}{2a^2 C B} = \Lambda$

R_{44}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{55}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{66}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{77}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{88}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{99}: $\frac{3\dot{a}\dot{b}}{a c^2 b[t]} + \frac{5\dot{b}^2}{c^2 b[t]^2} + \frac{\ddot{b}}{c^2 b[t]} = \Lambda$

R_{33} and R_{55} – R_{99} are redundant, the same as R_{22} and R_{44}, respectively.

R_{01} is an identity because $A(t,r) = c^2$.

R_{00} and R_{44} are two of the three equations we are seeking.

R_{11}–$2R_{22}$ can be rearranged to have only r terms on the left and only t-terms on the right, so both sides equal a constant that we call $-2c^2k$:

R_{11}–$2R_{22}$: $-\frac{2c^2}{r^2 C} + \frac{2c^2}{r^2 B} + \frac{2c^2 C'}{r C B} + \frac{c^2 (C')^2}{2C^2 B} = -a^2 c^2 \Lambda + 2\dot{a}^2 + a\ddot{a} + \frac{6a\dot{a}\dot{b}}{b[t]} = -2c^2 k$

$\Rightarrow$ (1) $\frac{2c^2}{r^2 B} + \frac{2c^2 C'}{r C B} + \frac{c^2 (C')^2}{2C^2 B} = -2c^2 k + \frac{2c^2}{r^2 C}$

and (2) $\frac{2\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{6\dot{a}\dot{b}}{a b[t]} = c^2 \Lambda - \frac{2c^2 k}{a[t]^2}$

Solving (1) for B yields the same result as in 4D:

$$B(r) = \frac{(2C + rC')^2}{4C\left(1 - k r^2 C\right)}.$$

Plugging this in, the equations become:

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} + \frac{6\,\ddot{b}}{b[t]} = c^2\,\Lambda$$

$$R_{11}: \quad \frac{2\,c^2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{6\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda$$

$$R_{22}: \quad \frac{2\,c^2\,k}{a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{6\,\dot{a}\,\dot{b}}{a\,b[t]} = c^2\,\Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{5\,\dot{b}^2}{b[t]^2} + \frac{\ddot{b}}{b[t]} = c^2\,\Lambda$$

R_{22} is redundant, and we have obtained the desired 3 ordinary differential equations.

Moreover, the Friedmann continuity equation is derived as an algebraic combination of these equations:

$$-\frac{1}{2}R_{00}+\frac{3}{2}R_{11}+3R_{44}: \quad \frac{3\,c^2\,k}{a^2} + \frac{3\,\dot{a}^2}{a^2} + \frac{18\,\dot{a}\,\dot{b}}{a\,b[t]} + \frac{15\,\dot{b}^2}{b[t]^2} = 4\,c^2\,\Lambda$$

Moduli Stabilization Problem

If the curled space is frozen ($\dot{b} = b = 0$), R_{44} reduces to $0 = c^2\Lambda$.
A static Calabi-Yau-sized internal space is therefore possible only if $\Lambda = 0$.
With a positive cosmological constant, the internal space must change size.